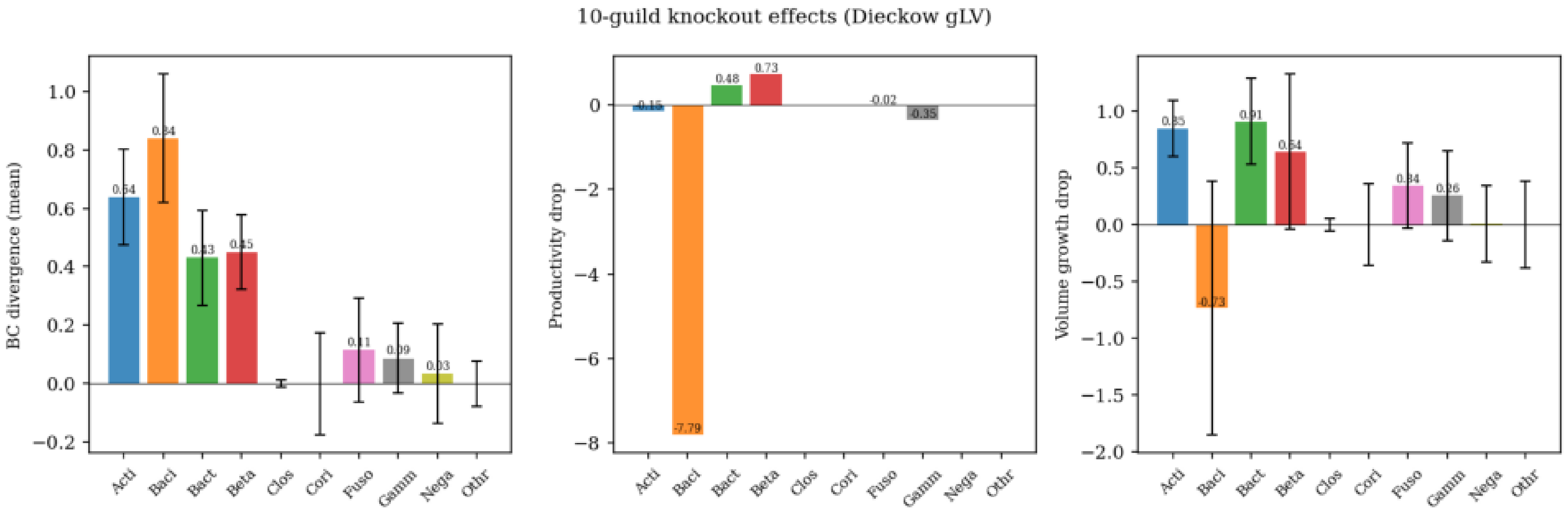
Guild-Level Dynamical Analysis

Keystone Analysis | **Network Centrality** | **Relapse Dynamics** | **Prevention Threshold** | **3D Phase Diagram**

Key question: What does the fitted gLV A matrix tell us about the long-term fate of the oral microbiome after dental treatment?

Guild-Level Keystone Analysis

Knockout experiment: remove one guild → simulate 21 d → measure BC shift / productivity / volume change



Key Findings

Compositional keystone:

Bacilli BC = 0.839

→ removal shifts community most

Competitive suppressor:

Baci vol_drop = −0.73

removing Baci → volume ↑

(other guilds fill the space)

Growth engine:

Actinobacteria +0.846

→ drives total biomass

Strongest interaction:

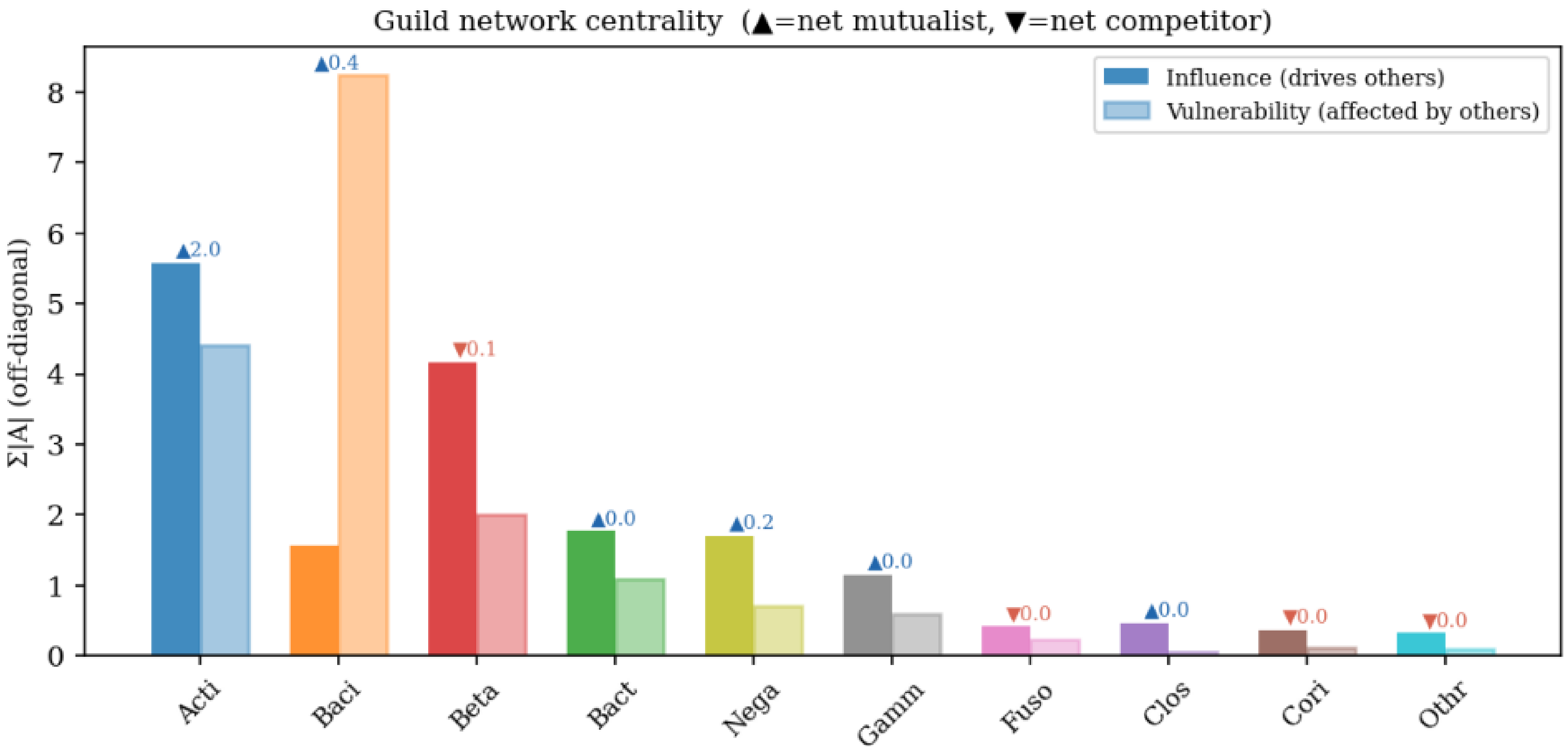
Acti → Baci A = +3.43

→ Acti stimulates Baci

Bacilli: highest BC divergence (0.839) + vol_drop = −0.73 → compositional keystone & competitive suppressor. | Actinobacteria: vol_drop = +0.846→ growth engine. | Strongest interaction: Acti → Baci A = +3.43.

Guild Network Centrality: Influence, Vulnerability, Net Role

Influence = Σ|A_ji| (column sum = how much guild i drives others) · Vulnerability = Σ|A_ij| (row sum = how much others drive guild i)



Network Roles

Mutualist hub:

Actinobacteria

influence = 5.58 (highest)

net effect = +2.02

→ drives all other guilds

Most regulated:

Bacilli

vulnerability = 8.24 (highest)

→ controlled by others

Net competitor:

Betaproteobacteria

net effect = −0.14

→ suppresses other guilds

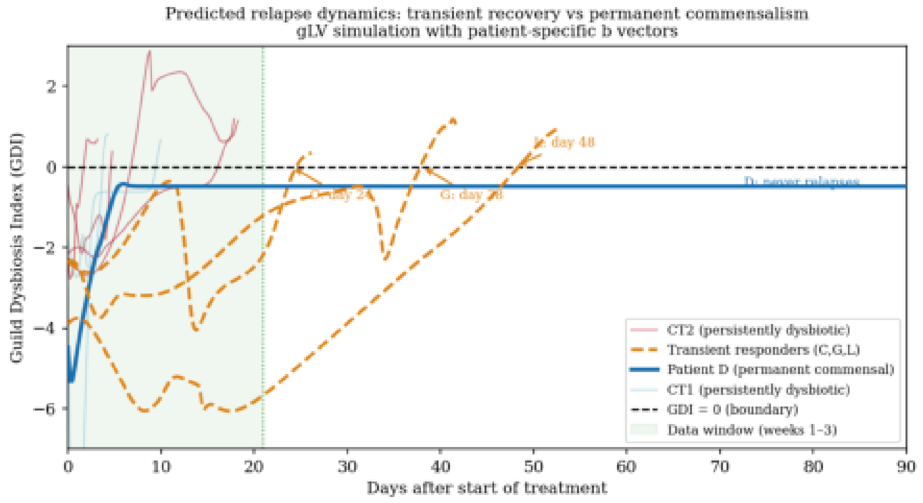
→ Acti–Baci–Beta triad

stabilises dysbiotic attractor

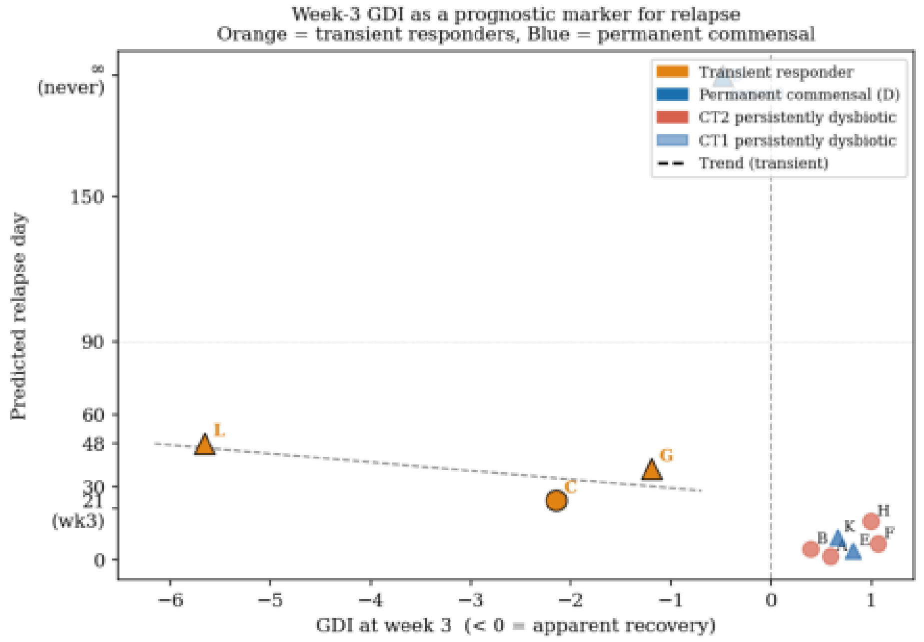
Actinobacteria: influence 5.58, net +2.02 → mutualist hub. Bacilli: vulnerability 8.24 → most regulated by others (compositional keystone confirmed). Betaproteobacteria: net−0.14 → only net competitor in the network.

Predicted Relapse Dynamics: Three Patient Archetypes

gLV simulation to t = 150 d with patient-specific b vectors · GDI = log(φ_dys) − log(φ_com) · GDI < 0 = commensal



(a) GDI trajectories to 90 d (per patient)



(b) Week-3 GDI as a prognostic marker for relapse

Three archetypes

① Persistent dysbiotic

A, B, E, F, H, K

GDI > 0 from day 1

② Transient responder

C → relapse day 24

G → relapse day 38

L → relapse day 48

commensal at wk-3,

then attractor pulls back

③ Permanent commensal

Patient D only

GDI stable < 0

Prognostic rule:

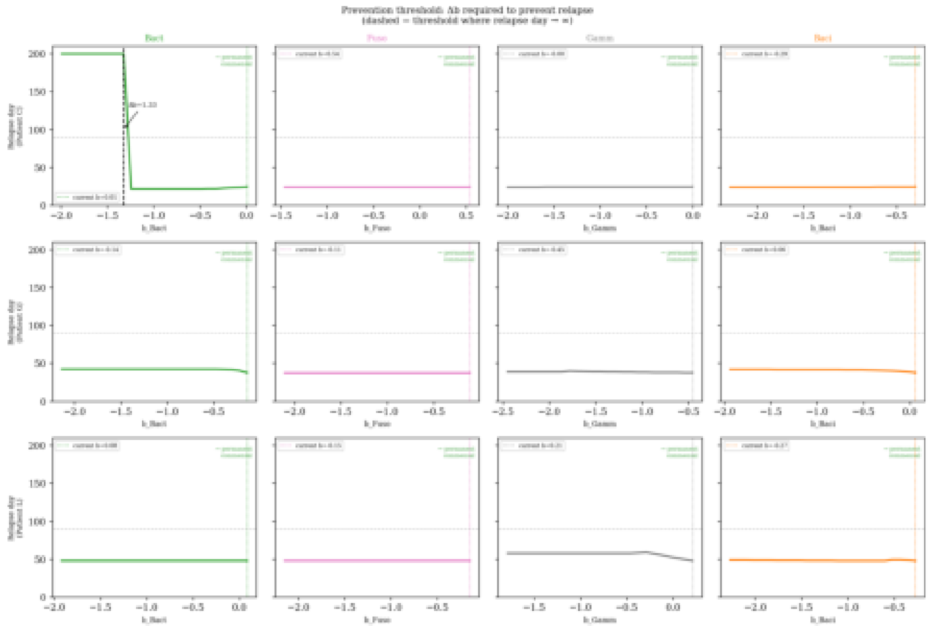
GDI(wk3) < −5 → >6wk

GDI(wk3) > −2 → <4wk

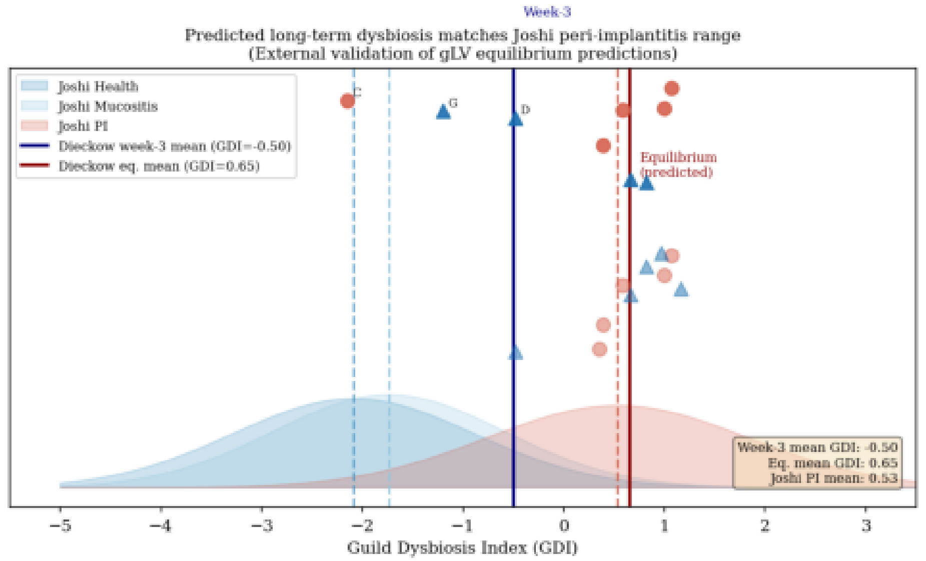
① Persistent dysbiotic (A,B,E,F,H,K): GDI > 0 throughout — gLV attractor dominates from day 1. ② Transient responders (C→24d, G→38d, L→48d): commensal at wk-3, relapse by day 50. ③ Permanent commensal (D only): uniquely stable b vector keeps GDI < 0 at t = 150d. → Week-3 GDI is a prognostic marker for relapse timing.

Prevention Threshold & External Validation

Left: min. Δb_i to prevent relapse (single-guild b sweep) · Right: model equilibrium GDI vs Joshi 2025 peri-implant cohort (N=127)



(a) Required Δb per guild for relapse prevention



(b) External validation: Joshi 2025 peri-implant cohort

Key Results

Prevention:

Patient C:

Δb_Bact = 1.33 sufficient

→ permanent commensal

Patients G & L:

no threshold found

→ deep dysbiotic attractor

Joshi validation:

Predicted eq. GDI:

+0.65 (model)

Joshi PI mean:

+0.53 (data)

9/10 patients match

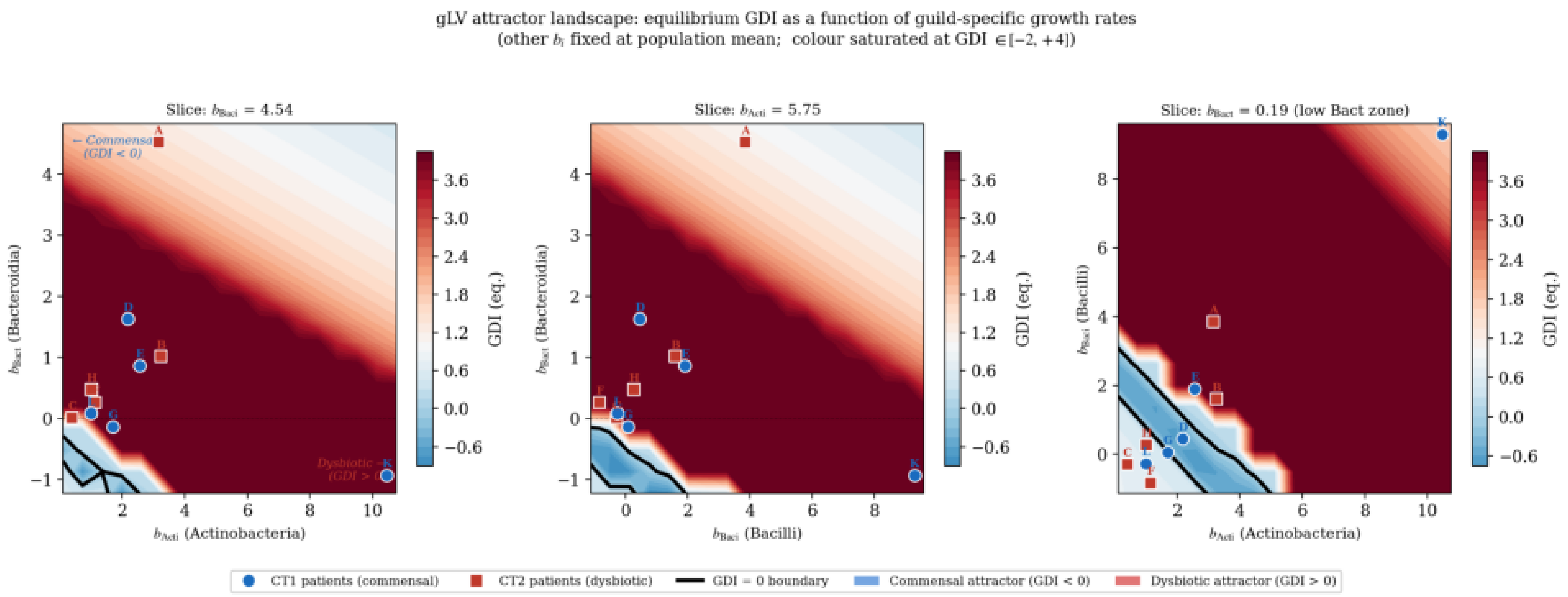
→ 3-wk data predicts

long-term dysbiosis

Prevention: Patient C convertible with Δb_Bact = 1.33; G & L require multi-guild change → deep attractor. | Joshi validation: predicted eq. GDI = +0.65 ≈ Joshi PI mean +0.53 (9/10 patients in PI range; Δ = 0.12 < Joshi SD = 1.30). → 3-week calibration extrapolates correctly to long-term peri-implant dysbiosis.

3D Attractor Landscape: GDI = 0 Boundary in Growth-Rate Space

18^3 grid · Acti × Baci × Bact · t = 80 d · 12-core parallel · Blue = commensal | Red = dysbiotic | Black contour = GDI = 0



Key Findings

Dysbiotic dominance:

98% of grid: GDI > 0

Only 2% commensal

Critical threshold:

b_Bact < ~1.5

→ commensal equilibrium

Bacteroidia = dominant

control variable

Patient positions:

CT1 (○): near boundary

CT2 patient A (b=4.5):

deep in dysbiotic zone

Clinical implication:

Must lower b_Bact, not

just change φ(0)

98% of parameter space is dysbiotic. Commensal region confined to b_Bact < ~1.5— Bacteroidia growth rate is the dominant control variable. CT1 patients with low b_Bact cluster near the boundary; CT2 patient A (b_Bact = 4.5) is deep in the dysbiotic zone. → Structural attractor robustness: initial-condition manipulation alone cannot achieve permanent commensalism.